

Can You Tell If a Bank Caller Is Real?

A plain-English summary of our phone-call voice research

September 11, 2026

What's the problem?

When you call a bank, you expect the person on the other end to be real. But today, AI can call in too — using a synthetic voice that sounds human. That's a security risk: fraudsters could use fake voices to trick customer service.

Our project asks a simple question: **listening to a phone recording, can we tell if the caller is a real person or a machine?**

What we did

We studied **353 real phone calls** in Mexican Spanish between bank customers and an automated bank agent. About **150 calls** were from real people. About **203** were from AI systems pretending to be callers (speech recognition + AI brain + synthetic voice).

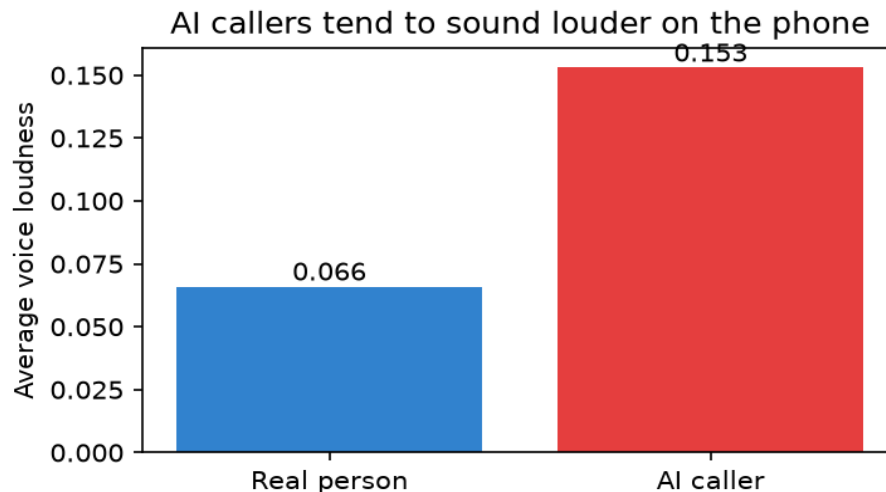
- Each call was recorded on two separate tracks — one for the caller, one for the bank agent.
- We built software to measure **how people talk** (timing, pauses, who speaks when).
- We also measured **how the voice sounds** (loudness, pitch changes, natural variation).
- We checked our work on calls the system had never seen before.

What we found

The biggest clue isn't what people say — it's how steady the fake voice sounds.

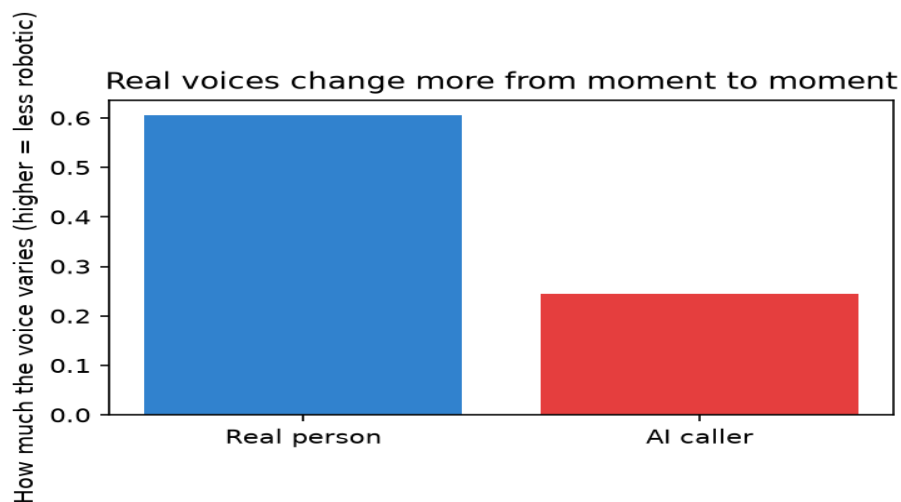
1. AI callers are louder

On average, synthetic callers come through the phone at roughly **twice the loudness** of real people. Think of it like a recording that's been volume-boosted.



2. Real voices are messier — and that's good

Real people don't sound the same every second. Their volume shifts, their tone wobbles, they pause unpredictably. AI voices are smoother and more **same-y** — like a well-produced audio clip on repeat.



3. AI callers talk more

Synthetic callers held the floor longer — about **26%** of the call vs **18%** for real people. Real calls had more time with the bank agent speaking.

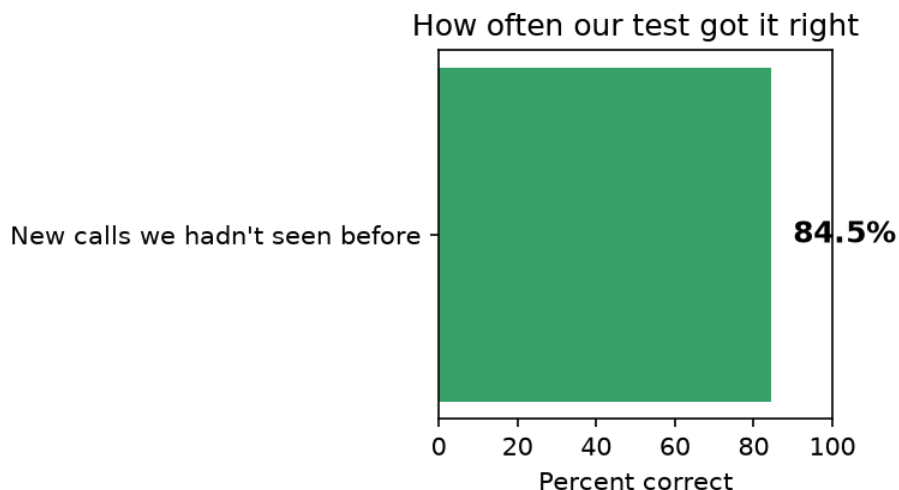
4. Real people react less predictably

When the bank agent stops talking, real callers don't always respond on the same beat. AI callers tend to jump in on a more regular schedule — but the **sound** of the voice was still the stronger giveaway overall.

In plain terms	What we saw
Fake voices	Louder, smoother, more uniform
Real voices	Quieter, more varied, less predictable
Who talks more	AI callers speak more; real callers get interrupted more

How well does it work?

Using just three sound-pattern measurements, our first automatic test correctly labeled callers about **85 out of 100 times** on phone calls it had never heard before.



That's promising for a first pass — but not perfect. A bank would need high confidence before blocking or flagging a caller, so more work remains.

Why this matters

Millions of people still bank by phone — especially where apps aren't an option. If synthetic voices become cheap and convincing, phone banking only stays trustworthy if the system can verify who is calling. This research is an early step toward that.

What comes next

- Build a live web tool banks could ping with a recording and get an answer back.
- Improve accuracy and confidence scores so the system knows when it's unsure.
- Test on more voices, accents, and phone conditions so it works in the real world.
- Combine sound analysis with conversation patterns for even better results.

This summary is based on exploratory research for a voice-security hackathon. It is not a finished product ready for production use.